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Authors	Yu, Qinkai;Zhao, He;Xu, Yanyu;Wang, Meng;Zhao, Yitian;Fu, Huazhu;Ye, Xujiong;Villavicencio, Aline;Lip, Gregory Y.H.;Zheng, Yalin;Meng, Yanda
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DiffGeo-AOR: Diffusion-Optimized Medical Grading via Geometric Priors enhanced Autoregressive Ordinal Regression

Qinkai Yu, He Zhao, Yanyu Xu, Meng Wang, Yitian Zhao, Huazhu Fu, Xujiong Ye, Aline Villavicencio, Gregory Y.H. Lip, Yalin Zheng, Yanda Meng

***Abstract*— Ordinal regression is well-known for leveraging the underlying inherent order between successive categories to obtain additional regularization beyond traditional probabilistic classification mechanism. However, there are challenges in real-world medical grading tasks: 1) The uneven distribution of disease severity levels, characterized by a long-tailed format, complicates the ordinal regression process. 2)The ambiguity in establishing disease severity thresholds introduces substantial challenges, rendering the ordinal regression framework susceptible to inter-class inconsistencies. To address the challenge, this work proposes *DiffGeo-AOR*, by introducing an autoregressive process to ordinal regression that operates directly on continuous global features, without any need for vector quantization. *DiffGeo-AOR* decomposes a K-class ordinal problem into K−1 conditional binary decision steps, enabling the model to explicitly infer whether the severity has crossed the next grade threshold at each step. We also introduces parameterized diffusion optimization to model conditional probability distributions, allowing continuous global features to be extracted and directly leveraged in the autoregressive process. In addition, we design a *FiLM*-gated Step-Aware Diffusion Conditioning Fusion that guides each step's decision based on both the current image representation and the previous soft prediction probabilities. Furthermore, we regularize the feature space with rank-anchored ordinal priors during training to facilitate stable convergence of the autoregressive module. *DiffGeo-AOR* consistently outperforms current state-of-the-art ordinal regression methods across both 2D and 3D medical grading tasks on three large-scale datasets. The implementation code is publicly available at `https://github.com/Qinkaiyu/DiffGeo-AOR`.**

***Index Terms*— Autoregressive Ordinal Regression, Diffusion-based Optimization, Medical Image Grading**

Q. Yu, and Y. Meng are with Bioengineering Program, Biomedical Sciences Division (BioMed), King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia. Y. Xu is with the Joint SDU-NTU Centre for Artificial Intelligence Research (C-FAIR), Shandong University, Shandong, China. M. Wang is with Centre for Innovation and Precision Eye Health, Yong Loo Lin School of Medicine, National University of Singapore, Singapore. H. Fu is with the Institute of High-Performance Computing, Agency for Science, Technology and Research, Singapore. Y. Zhao is with Ningbo Cixi Institute of Biomedical Engineering, Chinese Academy of Sciences, Cixi, China. Q.Yu, X. Ye, and A. Villavicencio are with the Computer Science Department, University of Exeter, Exeter, UK. H. Zhao and Y. Zheng are with the Eye and Vision Department, University of Liverpool, Liverpool, UK. Q.Yu and G. Lip are with Liverpool Centre for Cardiovascular Science, University of Liverpool, Liverpool, UK. Corresponding author: Yanda Meng (Yanda.Meng@kaust.edu.sa)

I. Introduction

Many clinical assessment tasks are naturally ordinal, including disease severity grading (e.g., diabetic retinopathy (DR) and age-related macular degeneration (AMD)) and medical image quality assessment (e.g., good, usable, and reject) [1]–[3]. For such tasks, ordinal regression is generally more suitable than flat multi-class classification because it explicitly models the monotonic relationship among categories and has shown clear practical benefits in medical imaging [4]–[10]. By incorporating ordinal structure, it can improve sample efficiency and yield more clinically coherent decision boundaries.

Despite these advantages, real-world medical grading remains challenging for existing ordinal regression methods. Two difficulties are particularly prominent. First, the label distribution is often highly long-tailed: mild or normal categories dominate the training data, whereas severe but clinically important categories are much rarer. This imbalance weakens the effective supervision received by high-severity samples and biases feature learning toward frequent classes. Second, the boundaries between adjacent grades are inherently ambiguous. In practice, neighboring severity levels often differ only subtly, and even expert annotations may vary due to subjective interpretation or temporal inconsistency [1]. As a result, medical ordinal prediction must address both long-tail imbalance and adjacent-boundary uncertainty.

We propose that autoregressive pipeline can address these two challenges. To be more specific, our autoregressive model reframes the single K-class ordinal problem into a sequence of $K-1$ simpler, conditional binary classification subproblems. For a task, such as Diabetic Retinopathy (DR) grading, with five levels of ordinal labels, the model does not perform a direct five-class classification. Instead, at the first step, it makes a binary decision to distinguish the 'No DR' class from all more severe classes. Conditioned on the outcome of the first step, it proceeds to the next, where its task is to distinguish the combined group of No DR, Mild from the remaining group of Moderate, Severe, Proliferative. This sequential process continues, with each step performing a simpler binary classification on a different grouping of the original classes. The final grade is determined by the sequence of these binary outcomes. This autoregressive decomposition yields two crucial advantages for medical grading. Firstly, it

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inherently addresses the long-tailed data distribution. Because determining a 'far' category (*e.g.*, 'proliferative DR') requires successfully traversing all preceding steps. This structure naturally amplifies the contribution of rare, high-severity samples to the overall loss optimization. Secondly, the mechanism effectively leverages the transitional nature of disease progression to resolve ambiguity. By making gradual and stepwise predictions that is conditioned on the previous outcomes, the model enhances its ability to discriminate at the subtle boundaries between adjacent grades. A formal theoretical analysis of these two effects is provided in Appendix I.

However, autoregressive ordinal regression paradigm remains largely unexplored. This is mainly because the autoregressive mechanism typically converts image patches into discrete tokens via patch-level features extractor backbones [11]. These features are then fed into a downstream decoder to capture contextual information across patches. This is the process commonly employed in object detection [12] or image segmentation [13] tasks. However, backbone networks used in image classification tasks are often specifically designed to extract continuous global features [14], which is contradictory to some extent. To address the above limitation, *Ord2Seq* [15] converts class labels into ordinal sequences and adopts a masked prediction strategy reminiscent of the *BERT* [16] training paradigm. However, such an pipeline fails to fully leverage pre-trained models, especially the powerful foundation models nowadays such as *RETFound* [17] and *MedSAM* [18]. A recent breakthrough by Li et al. [19] addressed this contradiction. They introduced a parameterized diffusion optimization framework that, for the first time, enabled autoregressive training directly on continuous features by using a diffusion process, eliminating the need for vector quantization. However, their contribution focused on the generative domain, and the potential of this powerful paradigm for the specific, discriminative task of ordinal regression remained unexplored.

In this work, we propose a novel autoregressive ordinal regression model called *DiffGeo-AOR*. It operates directly on continuous global features from the backbone and decomposes K classes into $K-1$ conditional binary decisions. Using medical severity grading task as one example, at the ordinal step j, the model needs to answer the question 'Did the severity crossed the j-th threshold given what we already decided at step $j-1$?'. Concretely, we adopt the pre-trained ViT [14] as the backbone and use `CLS` token as the global image features, then inject the previous steps' soft decision via a *FiLM* [20] gate to calibrate class-relevant channels. A learnable step embedding forms a single query that attends to such *FiLM*-modulated token, and a residual connection to the original `CLS` preserves image evidence across later steps during inference. To address the ambiguous category boundary challenges, we further reshape the feature distribution before decoding. Inspired by geometric order learning (GOL) [7], we introduce K learnable rank anchors that mark coarse ordinal locations in the embedding space and apply two complementary losses: (1) a directional order loss that aligns features with the forward rank direction, (2) a margin-based metric loss that separates adjacent ranks with an adaptive batch-wise margin. Different from GOL [7], our proposed anchors are only used during training to structure the space. And our model's inference process does not depend on geometric decisions nor anchor lookups. This avoids hardening fuzzy boundaries at test time and eliminates overhead, while still yielding features that progress monotonically with rank and remaining distinguishable across different category levels.

The contributions of this work are as follows:

- We introduce the first autoregressive framework for ordinal regression that is optimized via a parameterized diffusion process. Our novel paradigm operates directly on continuous global features, resolving the architectural conflict between autoregressive models and modern classification pipelines. This design is specifically tailored to address the core challenges of long-tailed distributions and inter-class ambiguity in medical grading.
- We design a step-aware conditional fusion module that dynamically integrates image evidence with the soft predictions from previous steps. This mechanism ensures a stable and context-aware autoregressive process by effectively regularizing the diffusion head with multi-step conditioning.
- We propose a novel geometric regularization scheme using rank-anchored ordinal priors that shapes the feature space exclusively during training. This strategy, which includes a directional loss and an EMA-adapted metric margin, stabilizes the convergence of the autoregressive module while adding zero inference cost. Because the priors are only used to constrain the feature distribution during training and are not utilized during inference.

Notably, this work is an extended version of [21], and we have made several different improvements and changes. First, we introduce the Rank-Anchored Ordinal Priors from Ordinal Geometry module to enforce an ordered geometric distribution of features in the embedding space, which facilitates more stable convergence during autoregressive training. Second, we redesign the original step-aware conditional fusion module—replacing the simple affine or cross-attention fusion with a FiLM layer-based cross-attention fusion mechanism. This redesigned module more effectively integrates image features with the historical predictions from previous steps, leveraging learnable step embeddings as queries and employing residual connections to preserve image evidence throughout subsequent inference stages. Third, compared with the MICCAI version that focused solely on diabetic retinopathy (DR) grading, we further evaluate our method on three different ordinal medical tasks to provide more comprehensive validation.

II. RELATED WORK

A. Ordinal Regression

Ordinal regression, also known as ordinal classification, extends beyond flat categorical learning by explicitly modeling the inherent order among classes, and is therefore more suitable for medical grading tasks. Early approaches [22] typically decompose a *K*-class ordinal problem into a set of simpler binary sub-tasks. A representative strategy [23] transforms the original problem into *K-1* binary threshold classification tasks, each corresponding to whether the sample exceeds a

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given ordinal boundary. However, these threshold predictors are usually optimized in parallel, and thus do not explicitly model the dependency among successive ordinal decisions. Moreover, independently trained heads may suffer from rank inconsistency, where the predicted probability of a higher rank paradoxically exceeds that of a lower one.

To alleviate this issue, *CORAL* [4] introduces a weight-sharing mechanism with a rank-consistent loss, and *CORN* [5] further improves ordinal head design through conditional training. While these methods are effective in enforcing output consistency, they still focus mainly on producing ordered threshold predictions. As a result, they do not explicitly address two challenges that are particularly important in medical grading: the subtle ambiguity between adjacent grades, and the long-tailed label distribution in which severe cases are much scarcer but clinically critical. In particular, parallel threshold heads do not model how previous ordinal decisions can help disambiguate subsequent adjacent boundaries, nor do they increase the effective supervision received by higher-severity samples.

Another line of work seeks to organize the feature space itself into an ordinal structure. One prominent family introduces ordered reference points or anchors. These anchors can be predefined [8], learnable [7], or dynamically sampled from the training set [24], [25]. During inference, the model determines the rank of a test sample by measuring its relative distance or relation to these references in the feature space. Representative methods such as *GoL* [7], QCN [26], and Moving Window Regression (MWR) [25] are effective in capturing ordinal geometry. However, such methods are fundamentally reference-based rather than decision-process-based: they infer ordinal relations through global feature comparisons, but do not explicitly model the progressive dependency among threshold decisions. In addition, methods such as MWR are mainly developed for fine-grained ordinal regression with many label levels (e.g., age estimation), and are less naturally suited to the small-K medical grading settings considered in this work.

A third family imposes ordinal structure through the training objective. By explicitly penalizing order violations or enforcing ordered similarity relationships, these methods guide the model to learn ordinally organized representations. This paradigm has been widely explored in vision-language learning, especially in CLIP [27]-based alignment frameworks [8]–[10], [28]–[30]. For example, CLIP-DR [10] introduces an ordinal-aware loss for diabetic retinopathy grading, NumCLIP [28] explores ordinal prompting through number supervision, RnC [29] enforces rank-consistent representation learning, and the more recent CLOC [30] provides a stronger contrastive benchmark for ordinal classification. Although these methods can improve the structure of the learned representation, they still do not explicitly formulate ordinal prediction as a progressive conditional decision process. In other words, they regularize the feature space or optimization objective, but do not model how one threshold decision should condition and inform the next.

Overall, existing ordinal regression methods still remain insufficient for medical grading. Parallel ordinal heads mainly focus on output consistency, reference-based methods emphasize global ordinal geometry, and loss-based methods regularize representation structure; however, none of them explicitly models a progressive conditional decision process that is tailored to the two key difficulties of medical grading, namely ambiguous boundaries between adjacent grades and long-tailed label distributions with scarce high-severity samples. To address this limitation, we introduce *DiffGeo-AOR*, which formulates ordinal prediction as a conditional step-wise decision process over continuous visual representations.

B. Autoregressive Modeling

Autoregressive models [31], [32] factorize a joint distribution into a sequence of one-dimensional conditional probabilities: $p(x) = \prod_i p(x_i|x_{<i})$, which now become a cornerstone of modern generative AI [32]–[34]. The common approach of this paradigm is to convert an image into a sequence of discrete visual tokens using techniques like vector quantization [35] or by simply treating image patches as tokens [36], [37]. This discrete sequence is then used for autoregressive decoding. Such a framework has been successfully employed in object detection [12] and segmentation [13] tasks. However, a fundamental conflict arises when attempting to apply this paradigm to standard image classification tasks. State-of-the-art backbone networks, such as the Vision Transformer (ViT) [14], are expressly designed to aggregate information into a single, continuous global feature vector (*e.g.*, the [CLS] token) for final prediction. This is inherently at odds with the discrete, sequential input format expected by classic autoregressive decoders.

Against this background, a few works have attempted to introduce sequence modeling into vision ordinal regression. Early efforts mainly relied on recurrent modules. For example, Zhang et al. [38] employ an LSTM to learn a spatial attention/localization mechanism for age-sensitive regions, rather than using recurrence as an ordinal classification head. Rajasekhar et al. [39] also introduce sequential modeling into the vision ordinal pipeline, but not in the form of conditional autoregressive threshold decoding over global image representations. Therefore, although recurrent components have been explored, prior sequence-based designs do not explicitly establish autoregressive decoding as an effective classifier head for continuous visual features in medical grading.

To overcome this chanllege, Ord2Seq [15] converts ordinal labels into a binary sequence and employs a BERT [16]-like masked prediction strategy. While innovative, this formulation is fundamentally different from causal autoregression: it treats ordinal prediction as masked sequence reconstruction, rather than as a progressive conditional decision process in which each step depends on the previously inferred ordinal state. As a result, it does not explicitly use prior ordinal decisions to reshape the current decision representation.

To address this, *Li et al.* [19] introduces a novel framework that enables autoregressive training directly on continuous global features by parameterizing the conditional distributions with a diffusion process, without image quantization. However, their contribution focused primarily on the generative domain,

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without extending the framework's utility to classification tasks. The potential of this paradigm remains largely unexplored for the specific domain of ordinal regression. Different from the aforementioned methods, our model is the first to propose an autoregressive ordinal regression paradigm optimized via parameterized diffusion.

III. Methodology

A. Preliminaries

For a classification task involving K classes with input images $I \in \mathcal{R}^{C\times H\times W}$, where C denotes the number of channels, H and W represent the image height, and the image width, respectively. Our autoregressive framework decomposes the ordinary process into *K-1* binary classification tasks. It is represented as $f : I \rightarrow y^{1:k-1}$, where $y^{1:k-1}$ denotes $\{y^1, y^2, ..., y^{K-1}\}$ and each $y^i \in \{0, 1\}$ is a binary variable. For example, the corresponding labels of 'Severe' in DR grading task (original label is 3) would be $\{1, 1, 1, 0\}$. To align with the autoregressive paradigm, we introduce a start token <BOS> and an end token <EOS> at the beginning and end positions. The complete label in our training process becomes $Y = \{\texttt{<BOS>}, Y^1, Y^2, \ldots, Y^4, \texttt{<EOS>}\}$. Thus, for the given label Y, our proposed autoregressive model reformulates the classic ordinal regression as the following optimization objective:

$$\max p(y^1, ..., y^4) = \max \prod_{i=1}^{4} p(y^i | y^1, ..., y^{i-1}), \qquad (1)$$

where $p(\cdot)$ represents the probability that the model makes the correct predictions.

B. Model Overview

Figure 1 illustrates the overall framework of DiffGeo-AOR. In the following, we briefly summarize the model architecture and the high-level training/inference workflow before presenting the detailed formulations of each component.

1) Model Architecture: The architecture contains three main components: a backbone encoder, a rank-anchored ordinal geometry module, and an autoregressive sequence decoder (ASD). The encoder extracts the global feature used by the downstream branches. During training, the geometry module regularizes the learned representation to respect ordinal progression. The ASD then performs conditional ordinal decoding, where the previous ordinal state and the current step index are fused with the global feature to construct the step-specific context for prediction. On top of this context, a lightweight diffusion decoder models the conditional output at each autoregressive step.

2) Framework Overview: As summarized in Algorithm 1, DiffGeo-AOR uses a unified autoregressive decoding framework for both training and inference. Given an input, the encoder first extracts the global feature used for step-wise decoding. During training, the rank-anchored ordinal geometry module additionally imposes geometric regularization on the learned representation. The autoregressive decoder is then unrolled from low to high thresholds: at each step, the current context is constructed from the global feature, the previous ordinal state, and the step index, and the diffusion module predicts the corresponding soft ordinal output. During inference, the same decoding process is applied recursively, but without the geometry losses. Detailed formulations of the rank-anchored priors, step-aware fusion, and diffusion-based conditional modeling are presented in Sections III-C, III-D, and III-E, respectively. A more detailed theoretical analysis of why the proposed autoregressive formulation helps alleviate long-tail imbalance and reduce adjacent-boundary ambiguity is provided in Appendix I.

Algorithm 1: Unified training and inference of DiffGeo-AOR for a single sample

Input: input I; ordinal label $y \in \{0, 1, \ldots, K-1\}$ for training
Output: $\mathcal{L}_{\text{total}}$ in training; predicted label $\hat{y}$ in inference

Training:
$\mathcal{X} \leftarrow \text{Enc}(I)$
if *ViT backbone* **then**
  $\mathcal{X} = [\mathbf{x}_{\text{CLS}}, \mathbf{x}_1, \ldots, \mathbf{x}_N]$
  $\text{GAP} \leftarrow \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i, \quad \mathbf{x} \leftarrow \mathbf{x}_{\text{CLS}}, \quad \mathbf{g} \leftarrow [\mathbf{x}; \text{GAP}]$
else
  $\mathbf{x} \leftarrow \mathcal{X}, \quad \mathbf{g} \leftarrow \mathbf{x}$
Compute $\mathcal{L}_{\text{ord}}$ and $\mathcal{L}_{\text{met}}$ from $\mathbf{g}$ and y
$\hat{\mathbf{y}}_{<1} \leftarrow \texttt{<BOS>}, \quad \mathcal{L}_{\text{diff}} \leftarrow 0$
for $j \leftarrow 1$ **to** $K-1$ **do**
  $y^j \leftarrow \mathbb{I}(y \geq j)$
  $\mathbf{C}_j \leftarrow \text{SDCF}(\mathbf{x}, \hat{\mathbf{y}}_{<j}, j)$
  $(\hat{y}^j, \mathcal{L}_{\text{diff}}^{(j)}) \leftarrow \text{DiffTrain}(\mathbf{C}_j, y^j)$
  $\hat{\mathbf{y}}_{<j+1} \leftarrow \hat{\mathbf{y}}_{<j} \oplus \hat{y}^j$
  $\mathcal{L}_{\text{diff}} \leftarrow \mathcal{L}_{\text{diff}} + \mathcal{L}_{\text{diff}}^{(j)}$
$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{diff}} + \lambda(\mathcal{L}_{\text{ord}} + \mathcal{L}_{\text{met}})$
Inference:
Compute $\mathbf{x}$ from $\text{Enc}(I)$ as in Training
$\hat{\mathbf{y}}_{<1} \leftarrow \texttt{<BOS>}$
for $j \leftarrow 1$ **to** $K-1$ **do**
  $\mathbf{C}_j \leftarrow \text{SDCF}(\mathbf{x}, \hat{\mathbf{y}}_{<j}, j)$
  $\hat{y}^j \leftarrow \text{DiffSample}(\mathbf{C}_j)$
  $\hat{\mathbf{y}}_{<j+1} \leftarrow \hat{\mathbf{y}}_{<j} \oplus \hat{y}^j$
for $j \leftarrow 1$ **to** $K-1$ **do**
  $\bar{y}^j \leftarrow \text{Round}(\hat{y}^j)$
$\hat{y} \leftarrow \sum_{j=1}^{K-1} \bar{y}^j$

C. Rank-Anchored Ordinal Priors from Ordinal Geometry

Inspired by Geometric Order Learning (*GOL*) [7], we structure the global feature distribution before autoregressive decoding so that the learned representations follow a consistent ordinal geometry.

Let $\mathbf{x}$ denote the unified global feature extracted by the encoder. For ViT-based backbones, $\mathbf{x} = \mathbf{x}_{\text{CLS}}$ is the class token

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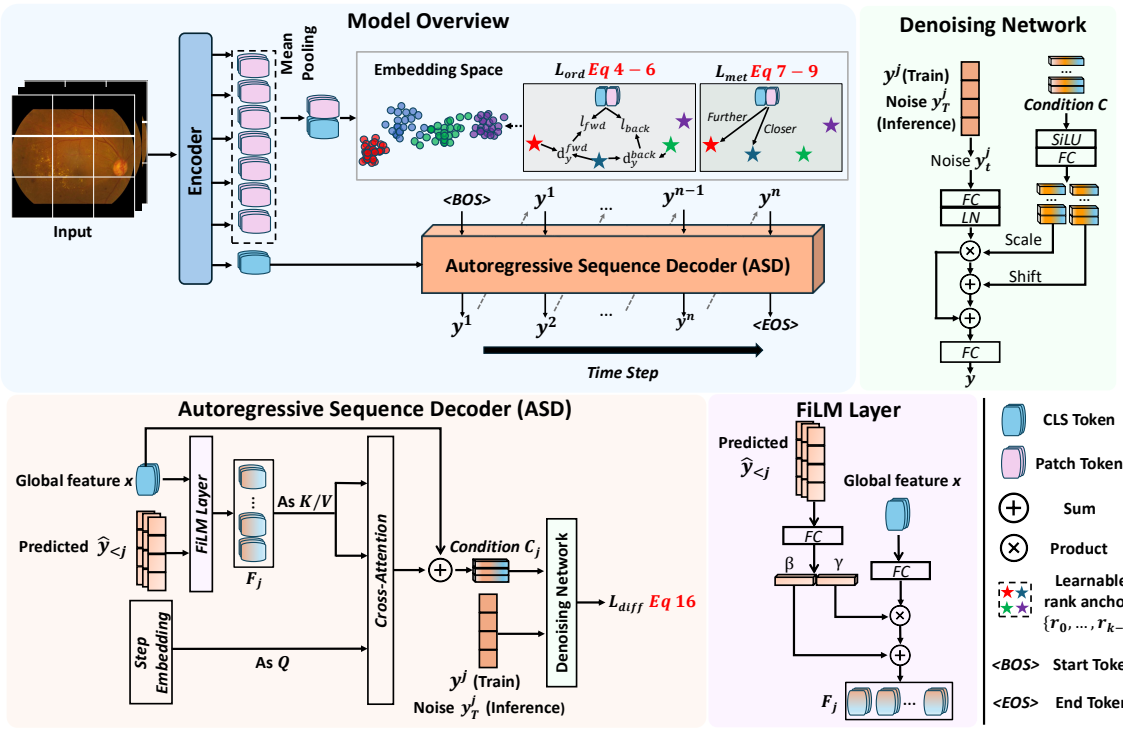


Fig. 1: Overview of the proposed *DiffGeo-AOR* framework. Our model architecture consists of two modules: Image Encoder and Autoregressive Sequence Decoder (*ASD*). The image encoder extracts patch-level features and a [CLS] token. They are then aggregated through mean pooling and concatenation operations to form a global representations g. We map the global representations g into an embedding space with K learnable rank anchors, where the proposed ordinal and metric constraints ($\mathcal{L}_{\text{ord}}$ and $\mathcal{L}_{\text{met}}$) are imposed. Specifically, the ASD takes the global feature [CLS] together with the autoregressive prefix $\hat{y}_{1:k-1}$ as input. The full prefix is used to encode the accumulated ordinal progression state before the current step, rather than relying only on the latest prediction. Then we apply *FiLM*-based conditioning and step embeddings operations, and integrates them via cross-attention. A denoising network predicts the clean target y from the noisy input $\tilde{y}_t$, which is optimized by the diffusion loss $\mathcal{L}_{\text{diff}}$. In addition, the *FiLM* layer generates γ and β from the previous steps' prediction $y^{1:k-1}$ via a *MLP* layer, which are then used in the Step-Aware Fusion for Diffusion Conditioning process. Notably, the architecture of the denoising network is referenced by the final layer designed in *DiT* [40]. The condition C is linearly transformed and decomposed into learnable scaling and shift. Then we merge it with the noise $\tilde{y}_t$ through residual connections for injecting conditional guidance into the denoising process.

representation; For CNN/R3D-style backbones, $\mathbf{x}$ denotes the globally pooled feature. Based on $\mathbf{x}$, we further construct a geometry feature $\mathbf{g}$ for ordinal-structure regularization.

For ViT-based backbones, let $\mathbf{X} = [\mathbf{x}, \mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}^{(N+1)\times d}$ denote the output token sequence, where N is the number of patches and d is the embedding dimension. We compute

$$\text{GAP} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i, \qquad \mathbf{g} = \big[\mathbf{x};\, \text{GAP}\big] \in \mathbb{R}^{2d}, \tag{2}$$

where $\mathbf{x} = \mathbf{x}_{\text{CLS}}$ in the ViT case. For CNN/R3D-style backbones, we simply use the pooled global feature itself as the geometry feature, i.e.,

$$\mathbf{g} = \mathbf{x}. \tag{3}$$

We introduce K learnable rank anchors, $\{\mathbf{r}_0, \ldots, \mathbf{r}_{K-1}\}$, to indicate coarse ordinal-level locations in the embedding space. For a sample with ground-truth label $y \in \{0, \ldots, K-1\}$, we normalize both the feature vector and the rank anchors: $\widehat{\mathbf{g}} = \mathbf{g}/\|\mathbf{g}\|_2$ and $\widehat{\mathbf{r}}_k = \mathbf{r}_k/\|\mathbf{r}_k\|_2$.

Directional Constraint ($\mathcal{L}_{\text{ord}}$) encourage features to align with the *local* forward direction between adjacent anchors. Specifically, the direction is defined only between immediately consecutive anchors, not between distant anchors. To formalize this, we first define the local directional vectors for a sample with ground-truth level y. Let $k^+ = \min(y+1, K-1)$ and $k^- = \max(y-1, 0)$, the forward direction vector $\mathbf{d}_y^{\text{fwd}}$ and the backward direction vector $\mathbf{d}_y^{\text{back}}$ as:

$$\mathbf{d}_y^{\text{fwd}} = \text{Norm}\big(\widehat{\mathbf{r}}_{k^+} - \widehat{\mathbf{r}}_y\big), \qquad \mathbf{d}_y^{\text{back}} = \text{Norm}\big(\widehat{\mathbf{r}}_y - \widehat{\mathbf{r}}_{k^-}\big), \tag{4}$$

with logits:

$$\ell_{\text{fwd}} = \langle \widehat{\mathbf{g}}, \mathbf{d}_y^{\text{fwd}} \rangle, \qquad \ell_{\text{back}} = \langle \widehat{\mathbf{g}}, \mathbf{d}_y^{\text{back}} \rangle. \tag{5}$$

Where Norm denotes the L2 normalization function and $\langle\cdot\rangle$ represent the dot product between two vectors. For the non-boundary levels ($1 \le y \le K-2$), we set the directional

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label to $y^{\text{dir}} = 1$ (forward), while boundary samples are downweighted or assigned $y^{\text{dir}} = 0$. The directional loss ($\mathcal{L}_{\text{ord}}$) is defined as:

$$\mathcal{L}_{\text{ord}} = \text{CE}\big([\ell_{\text{back}}, \ell_{\text{fwd}}],\, y^{\text{dir}}\big)\,. \tag{6}$$

Metric Separation ($\mathcal{L}_{\text{met}}$). To enhance class separability along the rank axis, we employ a triplet-style margin. The goal is to ensure that a sample's feature vector is closer to its correct rank anchor (the "positive") than to a more distant rank anchor (the "negative") by a specified margin. For a sample with ground-truth y, we select a positive anchor $\widehat{\mathbf{r}}_{a^+}$, and a negative anchor $\widehat{\mathbf{r}}_{a^-}$, where a^+ and a^- correspond to the positive and negative anchors indices respectively:

$$a^+ = y,\;\; a^- = \min(y+1,\, K-1), \tag{7}$$

and the distance to the anchor:

$$d^+ = \|\widehat{\mathbf{g}} - \widehat{\mathbf{r}}_{a^+}\|_2,\;\; d^- = \|\widehat{\mathbf{g}} - \widehat{\mathbf{r}}_{a^-}\|_2, \tag{8}$$

Thus the metric loss can be defined as:

$$\mathcal{L}_{\text{met}} = \max\big(0,\; m + d^+ - d^-\big). \tag{9}$$

The margin m adapts to batch difficulty using intra-class dispersion. Let σ_c be the mean feature standard deviation for class c in the current batch. We can define m as:

$$m_{\text{new}} = \beta \cdot \frac{1}{K}\sum_{c=0}^{K-1} \sigma_c, \qquad m \leftarrow \rho\, m + (1-\rho)\, m_{\text{new}}, \tag{10}$$

where $\beta \in [0.3, 0.5]$ for scaling and $\rho \in [0.8, 0.95]$ controls EMA (Exponential Moving Average) [41] smoothing.

Unlike GOL [7], our anchors are used exclusively during training to shape the feature distribution. we avoid GOL's [7] pairwise global batch ranking in favor of learnable anchors with EMA [41] adapted margins that track batch-wise intra-class dispersion. Such design can maintain separation signals commensurate with current feature noise levels (from long-tail and boundary-ambiguous data), preventing training-phase over- or under-separation while adding zero inference overhead.

D. Step-Aware Diffusion Conditioning Fusion

At the j-th ordinal step, the model should determine whether the current sample exceeds the j-th threshold conditioned on the previously inferred ordinal state. Therefore, the conditioning context for step j should contain two types of information: *(i)* the global image representation, and *(ii)* the step-specific ordinal context, including the previous soft prediction and the current step index. To achieve this, we construct a step-conditioned feature representation $\mathbf{C}_j$ from the global image feature and the autoregressive state.

For clarity, let $\hat{\mathbf{y}}_{<j} = \{\hat{y}^1, \ldots, \hat{y}^{j-1}\}$ denote the autoregressive prefix of soft predictions available before step j. We first inject this prefix through a FiLM-based modulation:

$$[\boldsymbol{\gamma}_j, \boldsymbol{\beta}_j] = h(\hat{\mathbf{y}}_{<j}) \in \mathbb{R}^{2d}, \qquad \mathbf{F}_j = \boldsymbol{\gamma}_j \odot FC(\mathbf{x}) + \boldsymbol{\beta}_j. \tag{11}$$

Here, $h(\cdot)$ is a lightweight MLP and $FC(\cdot)$ is a linear projection. Since the high-dimensional visual semantics are inherently captured by $\mathbf{x}$, the prior prediction $\hat{y}^{j-1}$ functions solely as a low-dimensional conditioning signal indicating the current ordinal state. Specifically, $h(\cdot)$ projects this scalar state into feature-wise affine parameters $(\boldsymbol{\gamma}_j, \boldsymbol{\beta}_j)$ to dynamically scale and shift the visual features. This mechanism closely mirrors adaptive conditioning techniques (e.g., AdaLN in Diffusion Transformers [40]), where low-dimensional signals guide network modulation. Through this FiLM-based formulation, the network adaptively reshapes the image representation based on the previous state, aligning it with the current threshold decision. We then encode the ordinal step index using a learnable embedding $e(j) \in \mathbb{R}^d$ to construct the step-specific query.

$$\mathbf{q}_j = W_q\, e(j). \tag{12}$$

We then compute a step-conditioned readout layer using a single-query cross-attention mechanism:

$$\mathbf{C}_j = \text{Attn}\big(\mathbf{q}_j,\; \mathbf{K}_j,\; \mathbf{V}_j\big) + \mathbf{x}, \tag{13}$$

where $\mathbf{K}_j = W_k\, \mathbf{F}_j$ represents the key features and $\mathbf{V}_j = W_v\, \mathbf{F}_j$ represents the value features. In our implementation, the key or value sequence has length$= 1$ (the modulated global token). So the attention mechanism reduces to a gated linear readout layer of $\mathbf{F}_j$ with output projection, stabilized by the residual $\mathbf{x}$. Such a designed residual connection preserves image evidence across late steps, preventing information washout as the chain grows.

E. Modeling Conditional Probabilities via Diffusion Process

Medical grading is inherently ordinal and progressive. While modern discriminative architectures excel at extracting global representations, their standard independent cross-entropy objectives struggle to capture the dynamic, step-wise conditional structure $p(y^j \mid C_j)$ required in our autoregressive formulation. To overcome this fundamental bottleneck, we introduce a parameterized diffusion process. Denoising diffusion networks are considered capable of modeling arbitrary distributions [42], [43]. A previous study [44] demonstrated the effectiveness of using diffusion models for modeling entropy, achieving remarkable performance in classification tasks. Recently, Li et al. [19] proposed a parameterized diffusion framework for autoregressive learning directly on continuous representations, bypassing the need for vector quantization. This formulation provides a natural mechanism to extend autoregressive modeling beyond discrete tokens.

We adopt this diffusion-based approach to handle the specific requirements of ordinal autoregressive decoding. Unlike standard discrete generation, our decoder operates on continuous global features extracted by the visual backbone. At each ordinal step j, the model estimates the conditional probability $p(y^j \mid C_j)$, where context C_j depends jointly on the global image representation and prior ordinal states. Standard cross-entropy optimization struggles to capture the dynamic conditional structure inherent in this autoregressive process. The diffusion framework overcomes this limitation by providing a tractable objective for modeling step-wise probabilities under dynamic visual contexts. We empirically

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validate this advantage over a standard cross-entropy baseline in our ablation study. Let y^j be the prediction at step j. We reformulate the optimization objective from Equation 1 to maximize $p(y^j \mid C_j)$, and detail the diffusion training and inference procedures below.

Training. For the j-th ordinal autoregressive step, let y_0^j denote the clean target and let y_t^j denote its noisy version at diffusion timestep t. We generate the noisy sample by

$$y_t^j = \sqrt{\bar{\alpha}_t}\, y_0^j + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \tag{14}$$

where $\epsilon \sim \mathcal{N}(0,1)$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is defined by the diffusion noise schedule. In our framework, the denoising network is parameterized to directly predict the clean signal rather than the added noise. Specifically, given the noisy sample y_t^j, diffusion timestep t, and step-specific context C_j, the network outputs

$$\hat{y}_0^j = f_\theta(y_t^j, t, C_j). \tag{15}$$

The conditional distribution $p(y^j \mid C_j)$ is then learned by minimizing the following diffusion-based objective:

$$L_{\text{diff}}(C_j, y_0^j) = \mathbb{E}_{\epsilon,t}\left[\|y_0^j - f_\theta(y_t^j, t, C_j)\|^2\right]. \tag{16}$$

During training, the autoregressive conditioning at step j is formed from the model's own previous soft prediction rather than the ground-truth label, so that the training and inference procedures remain consistent. In this way, the proposed autoregressive mechanism directly leverages continuous global features to model step-wise conditional probabilities in continuous space, without requiring an additional discrete autoregressive decoder over patch-level tokens.

Inference. For the j-th ordinal autoregressive step, let y_t^j denote the latent variable at diffusion timestep t, where $t \in \{0, \ldots, T\}$ and T is the total number of diffusion timesteps. The superscript j indexes the ordinal autoregressive step, while the subscript t indexes the diffusion timestep. Our goal is to recover the clean signal y_0^j conditioned on the step-specific context C_j. During inference, the reverse process starts from the terminal noisy state

$$y_T^j \sim \mathcal{N}(0,1). \tag{17}$$

In our framework, the diffusion decoder is parameterized to directly predict the denoised representation. Specifically, given the noisy sample y_t^j at timestep t, the network outputs

$$\hat{y}_0^j = f_\theta(y_t^j, t, C_j), \tag{18}$$

where $\hat{y}_0^j$ denotes the estimated clean signal for the j-th ordinal step. To improve inference efficiency, we adopt DDIM sampling during inference. The reverse update from timestep t to $t-1$ is written as

$$y_{t-1}^j = \sqrt{\bar{\alpha}_{t-1}}\, \hat{y}_0^j + \sqrt{1-\bar{\alpha}_{t-1}}\, \frac{y_t^j - \sqrt{\bar{\alpha}_t}\, \hat{y}_0^j}{\sqrt{1-\bar{\alpha}_t}}, \tag{19}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. After completing the reverse process, we obtain the final denoised output $\hat{y}_0^j$, which is converted into a soft probability

$$\hat{y}^j = \sigma(\hat{y}_0^j) \in [0,1], \tag{20}$$

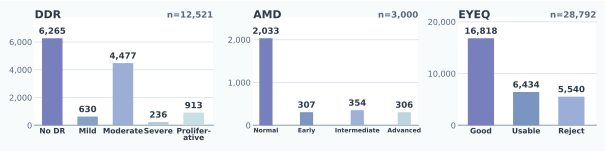


Fig. 2: **Visualization of long-tailed label distributions.** Class-frequency histograms for DDR Dataset, Harvard-30k AMD Dataset, and Eyeq Dataset. All three datasets exhibit a pronounced long-tailed pattern: the majority of samples concentrate in low-severity grades, while high-severity grades constitute a small minority.

where $\sigma(\cdot)$ denotes the sigmoid function. This soft prediction, rather than a binarized 0–1 decision, is used as the autoregressive conditioning signal for the next step. A hard binary decision is only used when converting the complete autoregressive sequence into the final ordinal label.

Finally, we use the overall loss function (L_{total}) to optimize the model parameters and the learnable rank anchors $\{\mathbf{r}_0, \ldots, \mathbf{r}_{k-1}\} \subset \mathbb{R}^{2d}$, which is given by:

$$L_{\text{total}} = L_{\text{diff}} + \lambda * (L_{\text{ord}} + L_{\text{met}}). \tag{21}$$

It is worth noting that L_{diff} updates the model encoder and the *ASD* parameters at each step. While L_{ord} and L_{met} are different that they only jointly optimizes the model encoder output $\mathbf{g}$ and reference point $\{\mathbf{r}_k\}$ once during a single batch training period. In this way, the feature space maintains monotonic progression along the ordinal direction while preserving inter-level distinguishability. This geometric prior ensures stable convergence of the diffusion head.

IV. Experiment

A. Experimental Settings

1) Datasets: We conducted extensive experiments on three large-scale datasets, covering three mainstream tasks for ordinal regression: diabetic retinopathy grading from fundus photographs, fundus image quality assessment, and age-related macular degeneration (AMD) grading from OCT scans. Compared with the earlier MICCAI version, which focused on multiple diabetic retinopathy grading benchmarks, the current study intentionally broadens the evaluation to a more general ordinal medical prediction setting across both tasks and modalities. Specifically, DDR serves as a representative 2D fundus grading benchmark, EyeQ evaluates ordinal fundus image quality assessment with a different clinical target, and Harvard-30k AMD tests 3D OCT-based severity grading. This design is intended to assess cross-task and cross-modality generalization, rather than optimize only for a single DR grading scenario. These datasets also exhibit diverse long-tailed distribution characteristics. Specifically, the dataset details of each task are shown below:

DDR: DDR [45] consists of 13,673 color fundus images captured using 42 different types of fundus cameras with a 45° field of view (FOV), primarily including Topcon D7000, Topcon TRC NW48, Nikon D5200, and Canon CR-2. The dataset provides image-level annotations for diabetic retinopathy (DR) grading. We used a cleaned subset provided by *Che*

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TABLE I: Comparison with recent state-of-the-art ordinal regression models across three datasets. Backbone-only baselines are included for reference. The best results are shown in **bold**, and the second-best results are underlined.

Model	*DDR* (DR Grading, ViT-B/16)				*EyeQ* (Quality Assessment, ViT-B/16)				*Harvard-30k* (AMD Grading, R3D-18)			
	Kappa	*F1*	*Sen*	*Spec*	*Kappa*	*F1*	*Sen*	*Spec*	*Kappa*	*F1*	*Sen*	*Spec*
Backbone-only baselines												
ViT-B/16 (baseline)	58.05	39.08	37.56	87.36	71.94	67.45	65.74	83.87	–	–	–	–
R3D-18 (baseline)	–	–	–	–	–	–	–	–	43.88	36.53	39.46	79.98
Ordinal regression heads												
CORAL [4]	64.37	44.58	43.04	88.33	86.03	78.15	79.34	90.41	48.77	40.50	40.32	81.21
CORN [5]	62.87	42.75	58.66	88.22	86.10	82.41	80.56	91.14	60.61	50.35	49.14	85.21
Loss-based ordinal methods												
PoEs [6]	59.99	37.81	36.40	87.52	84.47	78.16	79.00	85.88	50.65	34.15	34.86	83.42
CLIP-DR [10]	67.35	45.62	50.55	89.21	85.90	81.51	80.24	90.85	–	–	–	–
CLOC [30]	67.30	46.62	45.12	89.06	**90.01**	84.24	**86.10**	**93.65**	60.17	47.18	47.95	84.38
Reference-based ordinal methods												
GoL [7]	65.88	42.69	46.18	88.35	86.55	83.90	83.32	88.72	74.92	44.98	44.56	87.48
QCN [26]	47.57	46.08	50.81	88.59	63.45	76.78	77.75	88.97	74.58	52.99	52.02	87.34
Autoregressive / sequence-based ordinal methods												
Ord2Seq [15]	57.95	34.15	37.00	88.58	74.60	71.96	70.50	86.92	–	–	–	–
AOR-DR [21]	57.16±0.56	46.50±0.48	51.62±0.75	88.34±0.11	87.07±0.01	82.09±0.21	83.27±0.18	90.51±0.09	72.88±2.94	54.84±2.01	56.05±1.81	87.93±0.60
Ours	**68.78±0.45**	**47.06±0.65**	**51.85±0.64**	**89.28±0.13**	87.34±0.18	**84.56±0.26**	83.86±0.27	91.98±0.12	**76.38±1.77**	**57.60±1.29**	**56.42±1.28**	**88.09±0.45**

et al. [46] and adopted a five-level grading scheme: no DR, mild non-proliferative DR (NPDR), moderate NPDR, severe NPDR, and proliferative DR (PDR). We follow the same data split as defined in [46] for train, validation and test.

EyeQ: Eye-Quality (EyeQ) Assessment Dataset [3] is a re-annotated subset of the EyePACS dataset [47] designed for fundus image quality assessment. It contains 28,792 retinal images labeled with three levels of image ordinal quality: Good, Usable, and Reject. The dataset is split into a train & validation set with 12,543 images and a test set with 16,249 images.

Harvard-30k AMD: Harvard-30k AMD is a curated subset of the Harvard-30k dataset [48], which originally includes imaging data from 10,000 subjects diagnosed with age-related macular degeneration (AMD). We followed [49] and selected a high-quality subset consisting of 3,000 subjects with 3D optical coherence tomography (OCT) scans. Each OCT volume has a size of 128 × 200 × 200 (where 200 denotes the number of OCTs slices). There are four ordinal classes in this dataset: Normal, Early AMD, Intermediate AMD, and Advanced AMD, which represent a natural temporal progression of the disease.

We illustrate the long-tailed class distributions of the three adopted datasets in Figure 2.

2) Model: We compare DiffGeo-AOR with representative baselines under a unified experimental setting. For clarity, the compared methods in Table I are divided into two groups. The first group contains backbone-only reference models that use standard flat multi-class classification heads and are included only as non-ordinal references. The second group contains ordinal-specific methods, which differ in how the ordinal structure is modeled. These include ordinal-head methods (CORAL [4], CORN [5]), loss-based ordinal methods (PoEs [6], CLOC [30], CLIP-DR [10]), reference-based ordinal methods (GoL [7], QCN [26]), and sequence-based ordinal methods (Ord2Seq [15], AOR-DR [21], and our method). All compared methods are re-implemented using their publicly available code whenever possible.

3) Implement Details: To ensure fair comparisons, we adopt consistent backbone architectures and preprocessing strategies across all of the datasets used in this work. For the EyeQ and DDR datasets, all models use the same ViT-B/16 backbone [14], with input images resized to 224 × 224. For the Harvard-30k OCT dataset, we use the R3D-18 backbone [50], a 3D extension of ResNet-18 [51] designed for volumetric data, with input OCT volumes resized to 96 × 96 × 96.

To mitigate the randomness introduced by diffusion-based predictions, we perform five repeated inferences per sample and average the results. Image augmentations are limited to horizontal flipping. All models are trained with a batch size of 32 for 150 epochs. For the diffusion model, we use 1000 diffusion steps during training and 25 DDIM steps during inference to improve efficiency. The weighting coefficient λ in Eq. (21) is selected through validation-based hyperparameter search. We search λ in the range $[0.1, 0.5]$ and observe that the method is relatively stable within this interval. In all experiments, we use $\lambda = 0.2$, which provides the best overall trade-off across datasets.

4) Evaluation Metrics: We evaluate the model performance using four metrics: Cohen's Kappa (Kappa), Macro F1-Score (F1), Sensitivity (Sen), and Specificity (Spec). Kappa measures the label agreement corrected for chance, making it suitable for imbalanced data. Macro F1 averages the F1-score across classes, balancing precision and recall. Sensitivity measures the true positive rate, while Specificity measures the true negative rate.

B. Main Result

Table I presents a comprehensive comparison of our proposed *DiffGeo-AOR* model against recent state-of-the-art ordinal regression methods across three challenging datasets and tasks, respectively. Our model consistently demonstrates superior performance, outperforming all competitors on key metrics such as Kappa, F1, Sensitivity (Sen), and Specificity (Spec).

Backbone-only models lag significantly behind ordinal-specific methods, reaffirming the necessity of explicitly modeling ordinal structures for medical grading. Classic ordinal heads like CORAL [4] and CORN [5] remain competitive but struggle to fully resolve the severe class ambiguity and long-tailed distributions typical in these datasets. To address these issues, several distinct research lines have recently emerged,

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including loss-based approaches (e.g., CLOC [30], CLIP-DR [10]), reference-based models (e.g., GoL [7], QCN [26]), and sequence-based methods (e.g., Ord2Seq [15], AOR-DR [21]). While these advanced formulations provide much stronger baselines, DiffGeo-AOR consistently achieves the best overall trade-off across key clinical metrics.

On the highly imbalanced DDR dataset, DiffGeo-AOR establishes new state-of-the-art results across all metrics, achieving a Kappa of 68.78 and an F1 score of 47.06. Notably, our method reaches a Sensitivity of 51.85, highlighting its robustness in detecting minority severity classes compared to previous approaches like CLIP-DR (Kappa 67.35) and CLOC (F1 46.62). On the EyeQ dataset, our framework yields the highest F1 score (84.56) while maintaining highly competitive performance in Kappa (87.34) and Specificity (91.98), closely following CLOC [30] and demonstrating its versatility across different grading tasks.

Furthermore, on the large-scale Harvard-30k AMD dataset, which requires 3D volumetric encoding, DiffGeo-AOR substantially outperforms all applicable competitors, achieving a Kappa of 76.38 and an F1 of 57.60. This represents a significant margin over the second-best methods, GoL (Kappa 74.92) and AOR-DR (F1 54.84). It is also worth noting that several strong baselines, such as Ord2Seq [15] and CLIP-DR [10], cannot be applied to this dataset. Their reliance on 2D patch-based sequences or 2D image-text pre-training pipelines hinders their direct extension to 3D volumetric inputs—an architectural limitation our framework successfully overcomes.

C. Comparison with Vision-Language Pretraining

Given the increasing relevance of multimodal medical encoders, we further compare our framework with a representative vision-language model, namely MedCLIP [52], on the DDR and EyeQ datasets. For a fair comparison, we adopt the ViT-B/16 variant of MedCLIP and fine-tune its vision encoder on the target datasets. As shown in Table III, the fine-tuned MedCLIP baseline already provides strong performance, indicating that language-supervised medical pretraining yields highly competitive visual representations for medical grading. To further evaluate the extensibility of our framework, we replace the vanilla ViT-B/16 backbone in DiffGeo-AOR with the pretrained MedCLIP vision encoder and train the full model accordingly. DiffGeo-AOR with the MedCLIP encoder achieves further gains on several key metrics, particularly F1 score and sensitivity on DDR, and F1 score and specificity on EyeQ. These results suggest that our framework is not tied to a specific backbone, but is compatible with strong multimodal medical encoders and can further exploit their semantically enriched representations for ordinal prediction.

D. Ablation Study

To verify the effectiveness of the key components in DiffGeo-AOR, we conduct a granular ablation study, with results reported in Table II. All variants are evaluated under the same experimental setting for a fair comparison. The ablation is designed to isolate the contributions of four factors: step embedding, conditional fusion strategy, geometric regularization, and diffusion-based optimization. Since the extended framework changes the interaction between the autoregressive state, step-specific query, and feature fusion, a strict one-to-one module removal relative to the MICCAI 2025 AOR-DR formulation is not always well-defined. Therefore, instead of directly removing individual modules from the earlier design, we construct a progressive set of variants to isolate the contribution of each component. Specifically, Variant ❀(1) removes step-specific embeddings and uses a shared query across all autoregressive steps; Variant ❀(2) serves as the diffusion autoregressive backbone with step embeddings but without conditional fusion or geometric regularization; Variants ❀(3) and ❀(4) compare two fusion strategies, namely cross-attention and the proposed FiLM-based conditional fusion; Variants ❀(5) and ❀(6) further introduce $\mathcal{L}_{\text{ord}}$ and $\mathcal{L}_{\text{met}}$, respectively, on top of Variant ❀(4); and Variant ❀(7) replaces diffusion optimization with a standard cross-entropy objective under the same pipeline as the final model. The final DiffGeo-AOR combines step embeddings, FiLM-based conditional fusion, diffusion optimization, and both geometric regularizers.

Effect of Step Embedding. We first examine the role of step-specific embeddings by comparing ❀(1) with ❀(2). ❀(1) removes the step-specific query embedding and instead uses a single shared query across all autoregressive steps, while ❀(2) retains step-specific embeddings under the same diffusion autoregressive backbone. As shown in Table II, introducing step-specific embeddings consistently improves performance, indicating that explicitly distinguishing different ordinal decision steps is beneficial for autoregressive decoding. This suggests that the step embedding provides useful step-aware context and helps the decoder specialize to different threshold decisions.

Step-aware Diffusion Conditioning Fusion. We first evaluate the effect of the conditional fusion design in Table II. Comparing ❀(2) and ❀(4), introducing FiLM-based conditional fusion consistently improves performance across all three datasets, indicating that conditioning the current step on the autoregressive state is beneficial for ordinal prediction. We also compare FiLM with a cross-attention based fusion strategy in ❀(3). Although both strategies are effective, FiLM achieves overall stronger or more stable performance, especially on EyeQ and Harvard-30k AMD. This suggests that lightweight feature-wise modulation is a suitable way to inject step-conditional information into the decoder.

To further understand the role of FiLM-based conditional fusion, we visualize the feature representations before and after FiLM modulation at multiple autoregressive steps in Figure 3. FiLM performs feature-wise affine modulation conditioned on the previous autoregressive state, and its role is not to explicitly select a few visually identifiable channels, but to reshape the representation so that it becomes better aligned with the current threshold decision. As shown in Figure 3, the feature geometry changes consistently after modulation: samples associated with the same current-step prediction become more compact, while different prediction groups become better separated. This indicates that FiLM acts as a step-conditional

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TABLE II: Ablation study on key components: autoregressive paradigm (Diffusion vs. Cross Entropy Loss (CE)), step-aware conditioning fusion strategy, and geometry losses ($\mathcal{L}_{\text{ord}}$ and $\mathcal{L}_{\text{met}}$). All variants use the same backbones (ViT-B/16 for DDR & EyeQ; R3D-18 for Harvard-30k) and experimental settings. ✓ indicates that the component is enabled. CE-AR denotes step-wise cross-entropy autoregression.

Variant	Paradigm	Step embedding	Fusion Strategy	$\mathcal{L}_{\text{ord}}$	$\mathcal{L}_{\text{met}}$	DDR		EyeQ		Harvard-30k	
						Kappa	*F1*	*Kappa*	*F1*	*Kappa*	*F1*
❀(1)	Diff	✗	FiLM	✓	✓	62.29±1.02	42.82±0.97	80.35± 0.24	78.79±0.28	70.04±1.05	53.25±1.84
❀(2)	Diff	✓	None	✗	✗	64.70± 0.46	41.37±0.72	85.50±0.17	80.91±0.14	72.36 ±1.51	51.24±1.25
❀(3)	Diff	✓	Cross-Attn	✓	✓	64.98 ± 0.49	45.79 ± 0.58	87.12 ± 0.21	83.44 ± 0.22	73.92 ± 1.27	56.30 ± 1.63
❀(4)	Diff	✓	FiLM	✗	✗	65.95±0.45	43.95±0.70	85.94±0.22	82.02±0.28	71.05±1.46	53.98±1.94
❀(5)	Diff	✓	FiLM	✓	✗	66.95± 0.44	44.54± 0.88	86.48±0.17	82.44±0.27	72.63±1.45	54.62±1.31
❀(6)	Diff	✓	FiLM	✗	✓	64.97±0.72	42.89±1.01	86.25± 0.21	82.25±0.29	75.59±1.79	56.50±1.43
❀(7)	CE	✓	FiLM	✓	✓	62.20 ± 0	41.53 ± 0	83.78 ± 0	77.44 ± 0	0.00 ± 0	20.53 ± 0
DiffGeo-AOR	Diff	✓	FiLM	✓	✓	**68.78±0.45**	**47.06±0.65**	**87.34±0.18**	**84.56±0.26**	**76.38±1.77**	**57.60±1.29**

TABLE III: Performance comparison with the Vision-Language Model (MedCLIP [52]) on the DDR and EyeQ datasets. Both models utilize a ViT-B/16 architecture. For our method, we replace the vanilla backbone with the MedCLIP vision encoder.

Method	DDR				EyeQ			
	Kappa	**F1**	**Sen**	**Spec**	**Kappa**	**F1**	**Sen**	**Spec**
MedCLIP [52] (Fine-tuned)	**79.48**	56.43	54.27	**92.68**	**91.09**	88.25	**88.21**	94.20
DiffGeo-AOR (w/ MedCLIP encoder)	79.41±0.56	**58.76±0.42**	**58.03±0.12**	92.22±0.10	90.70±0.11	**88.90±0.09**	87.99±0.03	**94.27±0.05**

transformation that reorganizes the embedding space toward the current ordinal decision.

Contribution of Geometry Loss ($\mathcal{L}_{\text{ord}}$ and $\mathcal{L}_{\text{met}}$) We further evaluate the effects of the two geometric regularizers, $\mathcal{L}_{\text{ord}}$ and $\mathcal{L}_{\text{met}}$, on top of the diffusion autoregressive baseline with FiLM-based conditional fusion. As shown in Table II, both losses improve performance over ❀(4), indicating that geometric regularization is beneficial for ordinal representation learning. Specifically, $\mathcal{L}_{\text{ord}}$ encourages the model to preserve ordinal progression across severity levels, while $\mathcal{L}_{\text{met}}$ regularizes the relative geometric structure of the learned feature space. Their gains are complementary, and the final DiffGeo-AOR, which combines both regularizers, achieves the best overall performance. To further examine the effect of the proposed geometric priors, we compare the encoder features of AOR-DR and DiffGeo-AOR at matched training epochs in Figure 4. Since AOR-DR does not include $\mathcal{L}_{\text{ord}}$ or $\mathcal{L}_{\text{met}}$, this comparison provides a direct visualization of how the proposed geometric regularization reshapes the feature space during training.

Diffusion Optimization vs. Cross-Entropy Optimization. We finally assess whether the performance gain comes merely from the autoregressive decomposition or from the diffusion-based optimization itself. For this purpose, Variant (7) replaces the diffusion objective with a standard cross-entropy objective under the same pipeline as the final model. As shown in Table II, the cross-entropy counterpart yields consistently inferior results across all three datasets. This observation supports the motivation in Section III-E: diffusion-based optimization is better suited for modeling step-wise conditional distributions over continuous visual features than directly predicting discrete labels with a cross-entropy objective.

Inference Efficiency and DDIM Step Analysis. We further evaluate the practical inference efficiency of DiffGeo-AOR from two aspects: sensitivity to the number of DDIM sampling steps and overall computational cost compared with representative baselines. First, Table IV reports the sensitivity of the proposed method to the number of DDIM inference steps. For each configuration, the evaluation is repeated five times, and we report the mean and standard deviation. As the number of DDIM steps decreases, the inference speed improves substantially across all three datasets, while the predictive performance remains relatively stable. This indicates that the proposed framework is not highly sensitive to aggressive reduction of the sampling trajectory. Although fewer steps can occasionally yield comparable or slightly better results on individual metrics, 25 DDIM steps provide a favorable balance between efficiency and predictive stability across datasets. Therefore, we use 25 DDIM steps as the default inference configuration in all experiments. Second, we compare the model size and inference speed of DiffGeo-AOR against representative baselines in Table V. It should be noted that the proposed method does not employ a large denoising backbone such as a U-Net or DiT. Instead, the diffusion component is implemented as a lightweight MLP-based denoiser, while the computationally dominant backbone feature extraction is performed only once per image. Overall, these results suggest that DiffGeo-AOR achieves a reasonable efficiency–performance trade-off rather than incurring prohibitive computational cost.

V. Discussion and Conclusion

We propose DiffGeo-AOR, a diffusion-optimized autoregressive framework for medical ordinal regression. The central idea of our method is to reformulate ordinal prediction as a conditional step-wise decision process over continuous visual representations, rather than treating the task as either flat multi-class classification or a set of independent parallel threshold predictions. This design is particularly motivated by two challenges that are especially important in medical grading: long-tailed severity distributions and ambiguity between adjacent grades.

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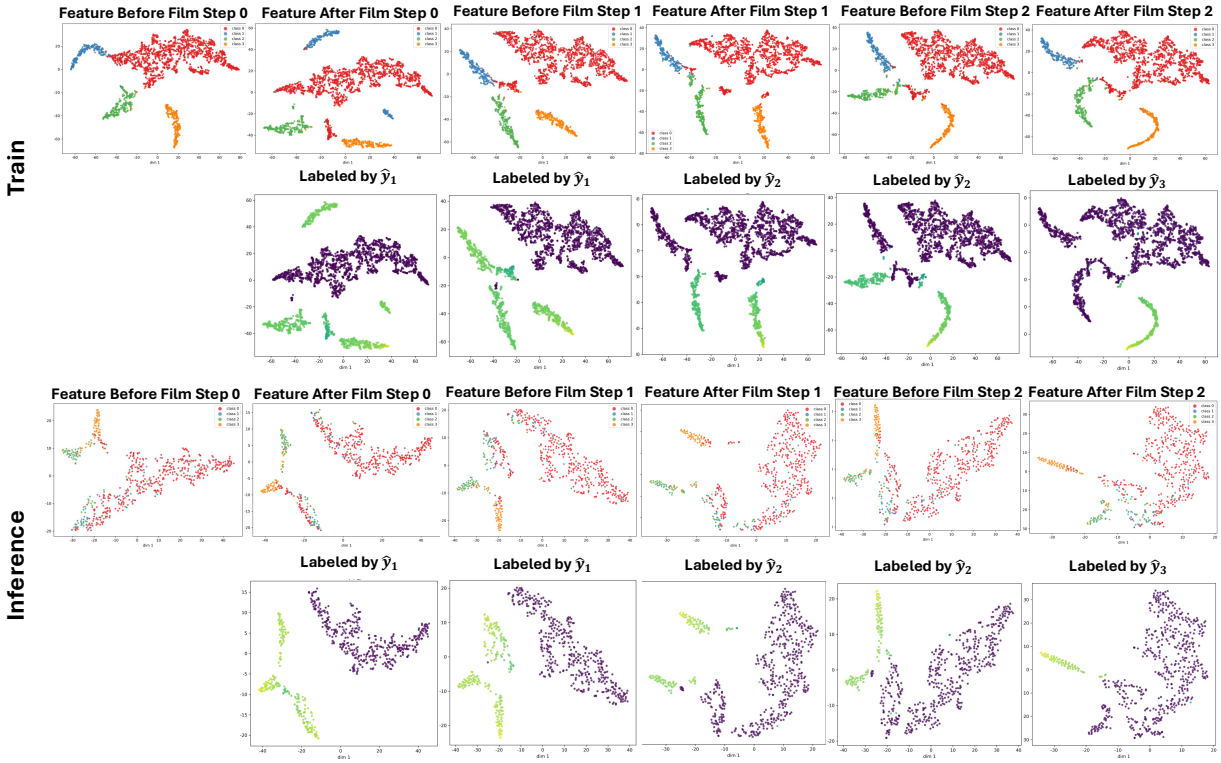


Fig. 3: **t-SNE visualization of representations before and after FiLM modulation on the Harvard-30k AMD dataset across autoregressive steps, shown for both training and inference.** Before FiLM, the feature distribution is still relatively mixed with respect to the current threshold decision and remains influenced by the previous autoregressive state. After FiLM, samples sharing the same current-step prediction form more compact groups, while different prediction groups become more clearly separated, indicating that FiLM reshapes the representation into a space better aligned with the current step-j decision. This trend is consistently observed across multiple autoregressive steps and in both training and inference. For **Step 0**, there is no valid $\hat{y}^{-1}$; thus the conditioning is initialized with a <BOS> token and no previous-step highlighting is shown.

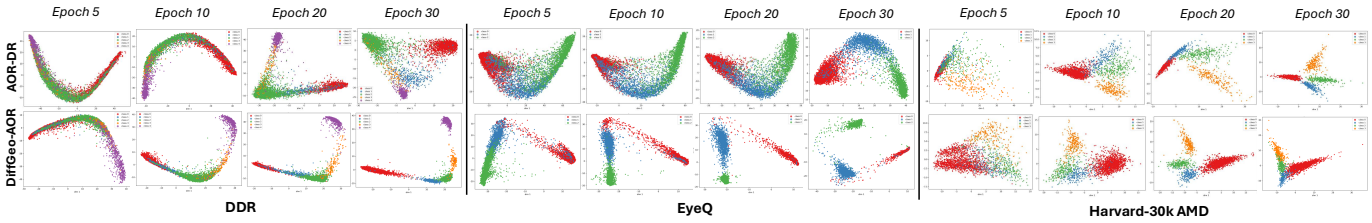


Fig. 4: Visualization of the effects of $\mathcal{L}_{\text{ord}}$ and $\mathcal{L}_{\text{met}}$ on representation learning. We extract encoder features before the autoregressive decoder at matched training epochs (Epochs 5, 10, 20, and 30) for AOR-DR and DiffGeo-AOR on DDR, EyeQ, and Harvard-30k AMD. Since AOR-DR does not include the proposed geometric regularizers, this comparison directly reveals their influence on feature evolution during training. Compared with AOR-DR, DiffGeo-AOR forms earlier, clearer, and more ordinally organized feature structures, indicating that the geometric priors facilitate faster convergence toward a more structured representation manifold.

This challenge-driven view also explains the roles of the main components in DiffGeo-AOR. The autoregressive formulation is used because medical severity prediction is inherently progressive: a high-grade prediction should be supported by the preceding threshold decisions, and later decisions should be conditioned on the ordinal state already inferred. However, applying this idea to medical image grading is not straightforward, since modern CNN and ViT backbones produce continuous global representations rather than discrete tokens. The diffusion-based objective is therefore used to optimize the step-wise conditional prediction directly in continuous feature space, without requiring visual tokenization or quantization. Meanwhile, the previous ordinal state is useful only when the learned representation is consistent with severity progression. The rank-anchored priors provide this training-time geometric structure, so that the autoregressive decoder operates on features aligned with the ordered label space. Together, the autoregressive path, diffusion objective, and rank-anchored

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TABLE IV: Sensitivity analysis of DDIM inference steps. We report the mean±standard deviation over five repeated evaluations. Reducing the number of DDIM steps substantially improves inference speed (FPS) while maintaining competitive predictive performance across all three datasets. Based on the efficiency–performance trade-off, 25 DDIM steps are used as the default inference configuration in the main experiments.

DDIM Steps	DDR					EyeQ					Harvard-30k				
	Kappa	F1	Sen	Spec	Fps	Kappa	F1	Sen	Spec	Fps	Kappa	F1	Sen	Spec	Fps
Steps = 100	68.33±0.42	46.88±0.51	51.70±0.65	89.23±0.13	103.73±0.24	87.36±0.12	84.60±0.17	83.93±0.18	92.01±0.06	156.40±0.91	75.49±0.93	57.73±1.47	56.59±1.44	88.36±0.38	122.16±0.83
Steps = 50	68.91±0.64	47.39±0.104	51.44±01.25	89.30±0.20	158.76±0.53	87.30±0.14	84.56±0.18	83.85±0.18	91.99±0.09	216.42±2.90	76.29±0.97	57.30±0.98	56.23±0.71	88.36±0.18	152.8±0.47
Steps = 40	68.73±0.58	46.89±1.04	51.85±1.33	89.29±0.13	177.66±0.81	87.28±0.04	84.50±0.05	83.82±0.05	91.96±0.03	234.61±2.39	75.68±0.41	58.44±1.26	57.44±1.07	88.29±0.27	159.44±0.65
Steps = 30	68.64±0.46	47.24±0.51,	51.98±0.54	89.27±0.12	202.27±0.38	87.36±0.14	84.63±0.22	83.93±0.23	92.02±0.09	258.52±1.41	75.38±1.64	57.78±1.51	56.45±1.25	88.33±0.41	165.24±0.09
Steps = 25	**68.78±0.45**	**47.06±0.65**	**51.85±0.64**	**89.28±0.13**	**216.27±0.63**	**87.34±0.18**	**84.56±0.26**	**83.86±0.27**	**91.98±0.12**	**270.00±1.62**	**76.38±1.77**	**57.60±1.29**	**56.42±1.28**	**88.09±0.45**	**171.28±0.24**
Steps = 20	68.57±0.49	46.70±0.75	51.67±0.66	89.21±0.10	233.46±1.37	87.30±00.13	84.58±0.14	83.86±00.17	91.98±0.07	283.47±0.47	76.81±0.81	58.39±1.54	57.21±1.48	88.48±0.23	176.24±0.09
Steps = 15	68.28±0.52	46.97±0.82	51.62±0.95	89.27±0.11	252.68±1.01	87.23±0.09	84.46±0.10	83.76±0.09	91.95±0.04	297.23±0.52	75.59±1.67	58.12±3.13	56.99±3.25	88.46±0.78	182.2±0.32
Steps = 10	68.76±0.48	47.03±0.48	51.81±0.51	89.27±0.10	276.47±0.36	87.40±0.09	84.66±0.13	83.96±0.11	92.03±0.05	313.03±0.26	76.41±0.96	58.27±0.77	57.02±0.78	88.42±0.22	189.2±0.31

TABLE V: Comparison of model size and inference efficiency on the DDR dataset using the ViT-B/16 backbone. All FPS measurements are obtained with batch size $= 64$. For DiffGeo-AOR, inference is performed with 25 DDIM sampling steps.

Model	Params (M)	FPS (img/s)↑
ViT-B/16	85.80	1270
CORAL	86.56	1276
CORN	86.57	1155
PoEs [6]	87.74	301
GoL [7]	85.69	77
QCN [26]	92.86	115
Ord2Seq	133.53	337
CLIP-DR	124.00	479
CLOC [30]	85.80	292
AOR-DR	107.10	55
DiffGeo-AOR (Ours)	112	216.27

priors make the step-wise ordinal prediction process feasible on continuous medical image features.

The experimental results suggest several key observations. First, the autoregressive formulation is beneficial for handling adjacent-boundary ambiguity. By conditioning each threshold decision on the previously inferred ordinal state, the model can progressively refine the current prediction instead of estimating all thresholds independently from the same feature. This makes the decision process better aligned with the progressive nature of disease severity. Second, the proposed rank-anchored ordinal priors further improve representation learning by encouraging ordinal progression in the feature space, which stabilizes training and enhances discrimination between nearby grades. Third, the comparison between diffusion-based optimization and the cross-entropy counterpart indicates that the performance gain does not arise solely from autoregressive decomposition itself, but also from the diffusion-based optimization strategy for modeling step-wise conditional predictions over continuous visual features.

Another important observation is that the proposed framework is not tied to a specific backbone. The additional comparison with MedCLIP suggests that DiffGeo-AOR is compatible with stronger multimodal medical encoders, rather than competing with them.

Monotonicity and Invalid Sequence. While the proposed decoder does not impose a hard monotonicity constraint during inference, invalid sequences are inherently suppressed by our ordinal encoding and training dynamics. As detailed in Appendix I, the scalar grade label deterministically maps to a valid monotone sequence. Thus, non-monotone targets are strictly absent from the training data. Consequently, once the autoregressive process predicts a zero state, the learned dynamics naturally drive all subsequent steps to zero, effectively eliminating non-monotone outputs in practice.

Limitations and Future Work. While DiffGeo-AOR establishes state-of-the-art predictive performance, the autoregressive diffusion process inherently introduces inference overhead compared to single-pass classifiers. Although this is substantially mitigated by DDIM sampling, further accelerating the sequential decoding trajectory remains an important direction. Future work will also explore integra ting explicit uncertainty calibration into the autoregressive loop to enhance interpretability and reliability for real-world clinical deployment.

Another limitation concerns severe distribution shifts. DiffGeo-AOR assumes that the learned ordinal geometry remains reasonably aligned with test images. If this geometry is substantially distorted, the previous soft prediction $\hat{y}_{j-1}$ may become miscalibrated, making the autoregressive conditioning unreliable and potentially propagating erroneous cues to later steps. This is a general robustness issue of learned medical image classifiers under strong acquisition-related shifts [53]–[55], rather than a limitation unique to DiffGeo-AOR. Our soft FiLM-gated conditioning reduces hard error propagation by treating $\hat{y}_{j-1}$ as a flexible cue, but it does not provide formal robustness guarantees under arbitrary out-of-distribution shifts.

Future work will also explore integrating explicit uncertainty calibration into the autoregressive loop to enhance interpretability and reliability for real-world clinical deployment.

Conclusion. *DiffGeo-AOR* provides a unified, problem-driven solution for medical ordinal grading. By synergizing conditional autoregressive decoding, diffusion-based optimization, and geometry-aware priors, the framework effectively resolves class imbalance and boundary ambiguity across diverse imaging modalities and clinical tasks.

APPENDIX I
ADDITIONAL THEORETICAL ANALYSIS OF THE AUTOREGRESSIVE FORMULATION

In this appendix, we provide additional theoretical discussion on why the proposed autoregressive formulation is beneficial for medical ordinal prediction. Specifically, we analyze two aspects: *(i)* how the autoregressive decoding process alleviates long-tailed supervision imbalance, and *(ii)* how conditioning on the previous ordinal state reduces adjacent-boundary uncertainty.

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A. Effective Supervision under Long-Tailed Ordinal Distributions

Consider an ordinal dataset with class labels $c \in \{0, 1, \ldots, K-1\}$ and class frequencies

$$N_0 > N_1 > \cdots > N_{K-1},$$

which represents a long-tailed distribution in which high-severity classes are progressively rarer. Each class label c is encoded into an ordinal binary target sequence according to

$$y^j = \begin{cases} 1, & j \le c, \\ 0, & j > c, \end{cases} \qquad j = 1, \ldots, K-1.$$

For example, in a 5-class setting, the corresponding ordinal targets are

$$\{0,0,0,0\},\ \{1,0,0,0\},\ \{1,1,0,0\},\ \{1,1,1,0\},\ \{1,1,1,1\}.$$

For each sample, the autoregressive objective can be written as

$$L(\mathbf{x}, y) = \sum_{j=1}^{K-1} \ell_j\big(\mathbf{C}_j, y^j\big), \tag{22}$$

where $\ell_j(\cdot)$ denotes the step-wise loss at ordinal step j, and $\mathbf{C}_j$ is the step-specific conditioning context constructed from the global image feature $\mathbf{x}$, the autoregressive prefix $\hat{\mathbf{y}}_{<j}$, and the current ordinal step j.

Theorem 1: Under the deterministic ordinal prior induced by the above encoding, the effective loss for a sample from class c can be approximated as

$$L(\mathbf{x}, y) = \sum_{j=1}^{K-1} \ell_j\big(\mathbf{C}_j, y^j\big) \approx \sum_{j=1}^{\min(c+1, K-1)} \ell_j\big(\mathbf{C}_j, y^j\big). \tag{23}$$

Consequently, higher-severity classes provide more effective learning steps per sample, partially compensating for their lower sample frequency without introducing explicit class weights.

Proof: By construction, the training set contains no non-monotone ordinal target sequences such as $\{0,1,0,0\}$. Therefore, the ordinal encoding induces the deterministic prior

$$\mathbb{P}(y^j = 1 \mid y^{j-1} = 0) \equiv 0. \tag{24}$$

That is, once the previous ordinal state has reached zero, the subsequent target states must remain zero.

After this feature-independent rule is learned, the corresponding conditional prediction rapidly saturates. In particular, for suffix steps following an already inactive state, we have

$$p_\theta^j(\mathbf{x}) = \mathbb{P}_\theta(y^j = 1 \mid \hat{\mathbf{y}}_{<j}, \mathbf{x}) \to 0,$$

when the prefix $\hat{\mathbf{y}}_{<j}$ has already entered the all-zero suffix regime. Hence, for suffix steps beyond the effective boundary, we may assume that there exists a small $\varepsilon > 0$ such that

$$p_\theta^j(\mathbf{x}) \le \varepsilon, \qquad j > \min(c+1, K-1).$$

Under this condition, the corresponding suffix loss becomes negligible:

$$\ell_j(\mathbf{C}_j, 0) \le O(\varepsilon) \approx 0, \qquad j > \min(c+1, K-1). \tag{25}$$

Therefore, after the deterministic ordinal prior is learned, the suffix steps contribute only marginally to the total objective, which leads to the approximation in Eq. (23).

The implication is that samples from higher-severity classes remain active for more autoregressive steps and therefore contribute more effective optimization signal per instance. In contrast, low-severity samples quickly terminate into the constant-zero suffix regime. Thus, the autoregressive formulation partially counteracts long-tail imbalance by increasing the effective supervision received by rare high-severity samples. ■

This analysis also explains why invalid non-monotone outputs are structurally suppressed. Since the optimal behavior after the first zero is the constant-zero suffix, patterns such as $0 \to 1$ are disfavored by the autoregressive optimization dynamics, even without imposing an explicit hard monotonicity constraint at decoding time.

B. Conditional Reduction of Adjacent-Boundary Uncertainty

We next analyze why the autoregressive conditioning mechanism is helpful for reducing adjacent-boundary ambiguity. At the j-th ordinal boundary, the prediction target is the threshold event

$$y^j = \mathbf{1}(c \ge j).$$

In a flat classifier, or in an independent parallel $K-1$ ordinal formulation, each threshold is predicted directly from the same image representation $\mathbf{x}$. In that case, the uncertainty of the j-th boundary decision is governed by the conditional entropy

$$H(y^j \mid \mathbf{x}).$$

In the proposed autoregressive framework, the j-th decision is additionally conditioned on the previous ordinal state. To first consider the ideal case, suppose the true previous label y^{j-1} is available. By the chain rule of entropy,

$$H(y^j \mid \mathbf{x}, y^{j-1}) = H(y^j \mid \mathbf{x}) - I(y^j; y^{j-1} \mid \mathbf{x}), \tag{26}$$

where $I(\cdot;\cdot \mid \cdot)$ denotes conditional mutual information. Since mutual information is always nonnegative, it follows that

$$H(y^j \mid \mathbf{x}, y^{j-1}) \le H(y^j \mid \mathbf{x}). \tag{27}$$

Therefore, conditioning on the previous ordinal state can only reduce, and never increase, the uncertainty of the current boundary decision.

This inequality provides a principled explanation for why autoregressive decoding is beneficial in medical grading. Adjacent grades are often visually similar, and their decision boundaries are therefore highly ambiguous. By conditioning the current threshold decision on the previously inferred ordinal state, the model receives additional state information that helps refine the current decision context and reduces ambiguity relative to independent threshold prediction.

In practice, our method does not condition on the true previous binary label during inference, but on the autoregressive prefix of previous soft predictions:

$$\mathbb{P}(y^j \mid \mathbf{x}, y^{j-1}) \quad \longrightarrow \quad \mathbb{P}(y^j \mid \mathbf{x}, \hat{\mathbf{y}}_{<j}), \tag{28}$$

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where $\hat{\mathbf{y}}_{<j} = \{\hat{y}^1, \ldots, \hat{y}^{j-1}\}$ summarizes the preceding soft ordinal states. Although $\hat{\mathbf{y}}_{<j}$ is only an approximation to the ideal conditioning variable, it still carries informative ordinal-state information and serves as a continuous summary of the current progression state. Consequently, the autoregressive conditioning process provides a principled mechanism for reducing adjacent-boundary ambiguity in practice.

Overall, the autoregressive formulation contributes in two complementary ways: it partially mitigates long-tail imbalance by allocating more effective learning steps to higher-severity samples, and it reduces adjacent-boundary uncertainty by conditioning each threshold decision on the previously inferred ordinal state.

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